Experiment # 4

'To Determine the Specific Resistance of a Wire Using a Meter bridge.

Theory :

In the arrangement as shown in Fig: 01 if X and R by the unknown and known resistance respectively and l be the distance of the null point measured from the left end A of the meter bridge, then by the principle of the Wheatstone's network we get

$$\frac{X}{R} = \frac{L}{(100-l)}$$

or

$$X = \frac{Rl}{(100-l)} \quad \ldots\ldots\ldots(1)$$

When the resistances X and R are interchanged, we get

$$\frac{R}{X} = \frac{L}{(100-l)}$$

or,

$$X = \frac{R(100-l)}{L} \quad \ldots\ldots\ldots(2)$$

If now L be the length of the experimental wire in centimeters then

$$X = \frac{\rho L}{\pi r^2}$$

or,

$$\rho = \frac{X\pi r^2}{L} \quad \ldots\ldots\ldots(3)$$

Where ρ is the specific resistance of the material of the wire and r is the radius of the cross-section of the wire.

Thus, ρ is may be determined after measuring X, r and L.

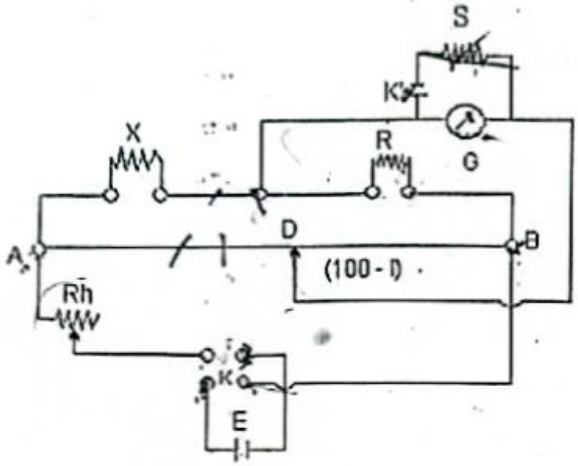

Fig : 01

Apparatus :

1) Meter bridge,
2) Leclanche's cell (E), DC ~~power~~ supply
3) Zero centre galvanometer (G),
4) Rheostat (Rh), Joke
5) Commutator (K),
6) Resistance box (R),
7) The specimen wire (X),
8) Connecting wires,
9) Screw- gauge etc.

ɟedure :

i) Make connections as shown in figure 1. Before putting in, fold one centimeter of the experimental wire at each end at right angles to the rest of it and put the folded portion with in the binding screws of the left gap. Insert 1 ohm resistance in the right gap. With 100 ohms or higher in the rheostat, move the sliding contact first to the left end and then to the right end of the bridge wire. If the deflections are on the opposite directions, the connections have been correctly made.

ii) Move the sliding contact along the bridge wire until the galvanometer deflection is almost zero. Null point is being approached.

iii) If the balance point seems to be far from the middle of the bridge wire, change the value of the resistance (R) in the right gap until the null point is brought very near the middle of the wire. Remove the galvanometer shunt by unplugging the key K' and find the null point accurately and note it. Reverse the current by changing the commutator plug and note the null point again. Take the mean of the two readings, thus eliminating thermoelectric effects. From the relation (1) calculate the value of X.

iv) Now interchange the position of X and R with R in the left gap. Find out the null point. Reverse the current and find out the null point again.

v) With two more known resistances, repeat the operations, every time reversing the current. Then calculate the mean value of X.

vi) Carefully determine the length L of the wire between two bends with ammeter scale.

vii) Measure the diameter (d) of the wire with a screw-gauge at several places with naturally perpendicular readings at each place.

ʒesult :

Length of the wire L = ------ cm.

Radius of the wire r = ------- cm.

Table I Reading for the balance point

Known resistance R in ohms	Positions of		Balance point (for l)			100-l cm	X ohms	Mean X ohms
	Unknown resistance x	Known resistance R	Direct	Reverse	Mean l cm			
1.	Left	Right						
	Right	Left						
2.	Left	Right						
	Right	Left						

Specific resistance of the material of the given wire is given by

$$\rho = \frac{X\pi r^2}{L}$$

= ---------- ohms-cm.

Discussion :

1) I observed if the null point is not far away from the middle.

2) I careful about tightness of every plugs in the resistance box R.

3) I took care to determine the radius (s) of the wire very accurately.

4) On reversing the current if the null point changes appreciably. The thermo-electric effect is too large. In such a case I closed the galvanometer circuit keeping the battery circuit open. The deflection of the galvanometer should be taken as the zero when looking for a null point.

5) E.M.F. of the cell was checked before starting the experiment.